

ECU Pulse

V 1.1

ECU Pulse is an open-source, browser-based engine diagnostics platform built to give the users real-time visibility into their engines.

Executive Summary

ECU Pulse is a real-time engine analytics platform that connects to any ELM327-compatible OBD2 adapter and displays live telemetry from an internal combustion engine. It is designed for riders, mechanics, and enthusiasts who want to understand their engine's health without paying for proprietary tools or dealer diagnostics.

The platform runs entirely in the browser. There is no server, no installation, and no account required. For users who do not have an OBD2 adapter, a built-in simulation engine generates realistic telemetry so the full dashboard experience can be evaluated before purchasing any hardware.

Every reading is visualized through stat cards, a tachometer arc, interactive charts, and a multi-axis health scoring system. All data can be exported to CSV for offline analysis or shared with a mechanic.

The Problem

Modern vehicles generate hundreds of data points every second. Engine RPM, exhaust temperature, air-fuel ratio, battery voltage, throttle position, oil pressure, and dozens of other parameters are constantly computed by the ECU. The owner's dashboard shows almost none of it.

What the driver sees is carefully curated. A check engine light tells you something is wrong. It does not tell you what, how bad, or how long it has been happening. The gap between what the engine knows and what the owner is told is enormous, and it is intentional.

How the Industry Keeps Owners in the Dark

- Proprietary scan tools. Manufacturer-grade diagnostics require tools that cost thousands of dollars and are licensed exclusively to authorized service centers. Independent mechanics are locked out. Owners are locked out.
- Selective data display. The instrument cluster shows speed, RPM, and fuel level. Everything else, including parameters that directly affect engine longevity, is hidden behind layers of abstraction or not exposed at all.
- Warranty leverage. Manufacturers use diagnostic data as leverage. If you modify your engine, use an aftermarket tuner, or even change your own oil, the data trail can be used to void warranty claims regardless of whether the modification actually caused the failure.
- Subscription-locked features. Several manufacturers now lock features like remote diagnostics, performance telemetry, and over-the-air updates behind monthly subscriptions. The data your car already generates is held hostage.
- Opaque error codes. OBD2 standardized basic fault codes, but manufacturers add proprietary PIDs and manufacturer-specific codes that generic scanners cannot read. The full picture remains inaccessible.

The average vehicle owner has no idea what condition their engine is in beyond what the dashboard warning lights tell them. Those lights are designed to protect the manufacturer, not the owner.

The Solution

ECU Pulse bridges the gap between the ECU and the vehicle owner. It reads live PIDs directly from the engine control unit through a standard OBD2 adapter and displays every available parameter in real time on a dashboard that is designed to be understood, not to confuse.

The platform does not replace your mechanic. It replaces the information asymmetry that lets dealerships charge for diagnostics that take five minutes on a laptop. When you can see your engine data yourself, you can make informed decisions about maintenance, modifications, and repairs.

What ECU Pulse Provides

- Real-time visibility into every sensor your ECU reports. Six primary stat cards, a tachometer arc, and four interactive chart panels update every simulation tick with live data.
- A health scoring system that evaluates five axes of engine condition: Engine, Thermal, Fueling, Electrical, and Mechanical. Each axis is computed from live sensor data using weighted heuristics.
- Full CSV logging. Every data point is timestamped and exported in a format that opens in Excel, Google Sheets, or any analysis tool. Share data with your mechanic without needing proprietary software.
- A physics-based simulation engine that generates realistic telemetry for users who want to evaluate the dashboard before buying an adapter. The simulation models five distinct riding modes with probability-weighted transitions and realistic sensor noise.
- Dark and light themes. Responsive layout that works on a laptop in the garage or a phone on the roadside. Zero dependencies. No accounts. No servers.

Core Features

Dashboard Layout

- Battery voltage with low and high threshold warnings.
- Engine temperature, exhaust gas temperature, and coolant tracking.
- Engine load percentage with real-time response.
- Vehicle speed from the ECU speed sensor.
- Air-fuel ratio monitoring for combustion efficiency.
- Tachometer arc with color-coded RPM zones.

Riding Mode Intelligence

- Five riding modes: Idle, City, Highway, Aggressive, Deceleration.
- Each mode defines a probability distribution across all sensor values.
- Auto-cycling transitions simulate real-world riding patterns.
- Manual override allows the user to pin a specific mode.
- A pie chart shows the distribution of modes across the current session.

Health Scoring

- Engine axis: RPM stress, engine load, throttle response.
- Thermal axis: EGT, engine temp, coolant temp, intake air temp.
- Fueling axis: AFR deviation from stoichiometric, MAP readings.
- Electrical axis: battery voltage stability.
- Mechanical axis: vibration analysis, oil pressure.
- Overall health score is a weighted blend of all five axes.

Data Import and Export

- One-click CSV export of all 18 telemetry parameters per sample.
- CSV import replays recorded sessions on the live dashboard.
- Re-importing the same file does not corrupt the data stream.
- Exported files include timestamp with and all sensor readings.

Technical Specifications

ECU Pulse is built with vanilla HTML, CSS, and JavaScript for the dashboard interface, and Python for OBD2 communication, simulation engine, and server-side bridging. There are no heavy frameworks or build tools.

Adapter Protocol:	ELM327 AT commands over serial
Supported PIDs:	RPM, Speed, Coolant Temp, Intake Air Temp, MAF, Throttle, Battery, Oil Pressure, Fuel Pressure, MAP, AFR, Vibration, Fuel Level, Engine Load, EGT
Simulation Model:	Physics-based with gaussian noise injection, probability-weighted mode transitions
Browser Support:	Chrome 89+, Edge 89+ (Web Serial). Simulation works in all modern browsers.

Getting Started

Simulation Mode

- Open ECU Pulse in any modern browser.
- Select Simulation on the mode selection screen.
- The dashboard begins generating realistic engine telemetry immediately.
- Switch between riding modes manually or let the auto-cycle run.
- Choose which graphs to be displayed by clicking on the legend of that graph.

Live OBD2 Connection

- Connect an ELM327 USB to your vehicle's OBD2 port.
- Turn the ignition to ON (engine can be off or running).
- Open the dashboard and select Live OBD2.
- Select the correct serial port when prompted by the browser.
- Data streams in real time as the ECU responds to PID requests.

The Vision

ECU Pulse is the first step toward a future where vehicle owners have full visibility into their engine's condition without relying on manufacturers, dealerships, or proprietary diagnostic chains.

Where We Are Taking This

- Let users define their own PIDs for vehicles with non-standard ECU configurations, including modified and tuned engines.
- Long-term health tracking. IndexedDB-backed history that tracks engine health trends over weeks and months, not just single sessions. Detect degradation before it becomes a failure.
- Predictive maintenance alerts. Machine learning models trained on aggregated sensor data to predict component failures before they happen. Not based on generic thresholds, but on YOUR engine's behavior over time.
- Multi-vehicle profiles. Switch between different cars or bikes with per-vehicle health history and baseline calibration.
- Integration with repair networks. Share health reports directly with trusted mechanics so they can diagnose issues remotely before the car even arrives at the shop.

Our goal is not to replace the mechanic. It is to give the vehicle owner enough information to have an honest conversation with one.

Current Limitations

- Browser Serial API is Chromium-only. Firefox and Safari do not support serial port access, so live OBD2 connections require Chrome or Edge.
- ELM327 adapter quality varies. Budget adapters may drop packets, return stale values, or fail to support certain PID modes.
- Simulation mode models typical engine behavior but does not replicate every edge case. It is a training and evaluation tool, not a digital twin.
- Health scoring uses heuristic thresholds. These are based on general IC engine tolerances, not manufacturer-specific calibration data.
- CSV import replays data sequentially. It does not reconstruct full chart history beyond the visible rolling window.

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